

CS3002E Compiler Design

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Lecture #25

Intermediate Code Generation - Conditionals, Boolean Expressions

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Grammar

$$S \rightarrow \text{if } (B) S \mid \text{if } (B) S \text{ else } S \mid \text{while } (B) S$$
$$B \rightarrow B \parallel B \mid B \&\& B \mid !B \mid (B) \mid E \text{ relop } E \mid \text{true} \mid \text{false}$$

if-else statement

Semantics of *if* (*B*) *S*₁ *else* *S*₂

if *B* evaluates to *true*, transfer control to *S*₁

if *B* evaluates to *false*, transfer control to *S*₂

The generated code should consists of code for *B*, followed by code for *S*₁, then code for *S*₂, with *goto* statement inserted at appropriate places according to the semantics.

Translation of *if-else* statement: example

if (a < b) small = a else small = b

if a < b goto L₁

goto L₂

L₁: small=a

goto L₃

L₂: small=b

L₃:

Translation of Boolean Expressions¹

Production	Semantic Rules
$B \rightarrow E_1 \text{ relop } E_2$	$B.\text{code} = E_1.\text{code} \parallel E_2.\text{code} \parallel$ $\text{gen}('if' E_1.\text{addr relop.op } E_2.\text{addr 'goto' } B.\text{true})$ $\text{gen}('goto' B.\text{false})$
$B \rightarrow \text{true}$	$B.\text{code} = \text{gen}('goto' B.\text{true})$
$B \rightarrow \text{false}$	$B.\text{code} = \text{gen}('goto' B.\text{false})$

$B.\text{true}$: label of the 3-address instruction to which control should transfer if B evaluates to true.

$B.\text{false}$: label of the 3-address instruction to which control should transfer if B evaluates to false.

¹Boolean expressions occurring in conditional control constructs. 

Translation of *if-else* statement

Production	Semantic Rules
$S \rightarrow \text{if } (B) S_1 \text{ else } S_2$	$B.true = newLabel()$ $B.false = newLabel()$ $S_1.next = S_2.next = S.next$ $S.code = B.code \parallel label(B.true)$ $\parallel S_1.code \parallel gen('goto' S.next)$ $\parallel label(B.false) \parallel S_2.code$

newLabel(): creates and returns a new label in each invocation.

label(L): attaches label *L* to the next 3-address instruction to be generated.

S.next (inherited attribute): stores the address (label) of the instruction that should immediately follow *S*.

Translation of *if* statement

Production	Semantic Rules
$S \rightarrow \text{if } (B) S_1$	$B.true = newLabel()$ $B.false = S_1.next = S.next$ $S.code = B.code \parallel label(B.true) \parallel S_1.code$

Exercise

Write the 3-address code generated for the following instructions, as per the given SDD:

if ($x < y$) $small = x$ else $small = y$

if ($true$) $small = x$ else $small = y$

References

References:

- Aho A.V., Lam M.S., Sethi R., and Ullman J.D. Compilers: Principles, Techniques, and Tools (ALSU). Pearson Education, 2007.

Further reading:

- ALSU Section 6.6